Auburn Governance Report

Reflection on $\mathbb{T}^3$ Subcritical Energy Estimates for the
3D Incompressible Navier–Stokes Equations

A Research Pamphlet Documenting the Technical Requirements,
Design Decisions, and Coverage of the Angular Cancellation Programme

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Purpose. This document is a technical companion to the manuscript *The Angular Cancellation Lemma: Machine-Verified Geometric Structure of Incompressible Triadic Transfer on* $\mathbb{I}^3$. It does not restate the results. It traces, in systematic detail, the landscape of requirements identified by the research community for progress on the 3D incompressible Navier–Stokes regularity problem, and documents how each requirement was addressed — or explicitly scoped as open — in the Angular Cancellation programme. The reader is invited to evaluate the coverage independently.

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1 The Supercriticality Barrier

1.1 What Was Required

In 2016, Tao established that the 3D Navier–Stokes regularity problem cannot be resolved by any argument relying solely on the energy identity, Sobolev embedding, and abstract bilinear estimates. Specifically, there exists an averaged bilinear operator $\tilde{B}(u, u)$ that preserves the energy cancellation $\langle \tilde{B}(u, u), u\rangle = 0$, obeys all standard function-space bounds, and yet admits finite-time blowup.

Any viable approach must therefore exploit **specific structural properties of the true Navier–Stokes nonlinearity** that are not shared by the class of averaged operators.

The follow-up work (2016) further narrowed the irreducible structural feature: a modified Euler equation preserving the energy identity, vorticity equation, circulation theorem, helicity conservation, and transport of vortex lines *still* admits blowup when the Biot–Savart law is modified. This implies that the **specific form of the Biot–Savart kernel**, encoding Euclidean geometry, is the irreducible distinguishing feature.

1.2 What Was Addressed

The Angular Cancellation Lemma identifies a geometric mechanism that is destroyed by the averaging process. The mechanism operates through three ingredients:

1. **Incompressibility orthogonality.** The divergence-free condition forces $\hat{u}(k) \perp k$ for every Fourier mode. This constraint is specific to the true Navier–Stokes nonlinearity. The averaged operator replaces the Leray projector with rotated and multiplied copies, destroying the rigid perpendicularity.

2. **Lattice shell geometry.** On the integer lattice $\mathbb{Z}^3$, the resonant set for local triadic interactions within a unit-width shell has $O(k_j)$ points (a one-dimensional annular band), not $O(k_j^2)$ (the full shell surface). This dimensional reduction is a property of the discrete torus $\mathbb{T}^3$ and its lattice spectrum; it does not survive averaging over rotations.

3. **First-harmonic oscillation.** The Leray projector forces the azimuthal dependence of the coupling kernel to be a pure first-harmonic cosine $B(\phi) = R(\theta)\cos(\phi - \phi_0)$, whose zeroth Fourier coefficient vanishes exactly. This structure is determined by the specific form of the Leray projector $\mathcal{P}_k = I - \hat{k} \otimes \hat{k}$, which is the Fourier-space manifestation of the Biot–Savart law.

Each ingredient fails under Tao's averaging: ingredient (1) because spatial rotations scramble the perpendicularity; ingredient (2) because rotations destroy the lattice structure; ingredient (3) because the Leray projector's specific quadratic dependence on $\hat{k}$ is replaced by pseudodifferential operators.

The combined mechanism yields a deterministic $k_j^{1/2}$ saving in the shell-local vortex stretching bound, improving the standard estimate from $|VS_j| \leq Ck_j^4 E_j^{3/2}$ to $|VS_j| \leq C_{\mathrm{ACL}} k_j^{7/2} E_j^{3/2}$.

1.3 Explicit Tao Test

The manuscript states and verifies the Tao test as a formal criterion (Corollary 4.4): any result that "passes the Tao test" must use structural properties of the NS bilinear form $B(u, u) =$

$\mathcal{P}[(u \cdot \nabla)u]$ not shared by all operators satisfying the same abstract estimates. The ACL passes this test because:

- The transverse-set dimensional reduction uses $\hat{u}(k) \perp k$ (absent from averaged systems).
- The bounded geometric factor uses the specific Leray projector (absent from averaged systems).
- The collinear counting bound uses the integer lattice geometry (absent from averaged systems).

2 The Lattice Geometry Frontier

2.1 What Was Required

The research literature identifies the arithmetic structure of $\mathbb{Z}^3$ as an "unexploited frontier" for Navier–Stokes analysis. While the Fourier spectrum on $\mathbb{T}^3$ is discrete, no published work prior to the present programme systematically exploited this discreteness to improve nonlinear estimates.

2.2 What Was Addressed

The entire Angular Cancellation programme is built on a lattice-first methodology. The key lattice-geometric results are:

1. **Unit-width shell decomposition.** The frequency space is decomposed into shells $S_j = \{k \in \mathbb{Z}^3 : j \leq |k| < j + 1\}$ of thickness equal to the lattice spacing. This is the intrinsic resolution of $\mathbb{Z}^3$; thinner shells would contain gaps, thicker shells would destroy the counting arguments.
2. **Resonant set counting.** For each output mode $k \in S_j$, the resonant input modes p satisfying $|p|, |k-p| \in S_j$ lie on a thin annular band with $\Theta(k_j)$ lattice points (Lemma 4.22). This is an arithmetic fact about $\mathbb{Z}^3$, not a continuum approximation.
3. **Transverse-collinear partition.** The resonant set is decomposed into a transverse subset ($|p_\perp| \geq c_0|k|$, containing $O(k_j)$ points) and a collinear subset ($|p_\perp| < c_0|k|$, containing $O(c_0 k_j)$ points). The partition constant c_0 is fixed universally, independent of shell index.
4. **Lattice equidistribution.** The passage from discrete sums to continuum integrals is controlled by the Duke–Linnik equidistribution theorem, producing an $O(k_j^{-1/2})$ error for smooth test functions on thin spherical shells.
5. **Collinear counting uniformity.** The collinear tube counting bound $\#\mathcal{C}^{\text{coll}} \leq C c_0 k_j$ holds uniformly in the direction of k, including for non-primitive lattice vectors where the covolume trades off against plane density.

3 The Two-Derivative Gap

3.1 What Was Required

The core technical obstacle in the regularity problem is the two-derivative gap: the vortex stretching term in the enstrophy equation involves three spatial derivatives of u, while viscous dissipation controls only two. Via Sobolev interpolation, this produces a super-quadratic ODE $d\Omega/dt \leq C\Omega^3/\nu^3$ that can blow up in finite time.

This pattern repeats at every level of the Sobolev hierarchy: the stretching term at the H^k level involves $(2k+1)$ derivatives, while dissipation controls $2k$. No finite chain of energy-type estimates closes by standard methods.

3.2 What Was Addressed

The manuscript does not attempt to close the two-derivative gap by brute-force derivative counting. Instead, it renders the gap irrelevant through structural decomposition of the nonlinearity.

The argument proceeds through a sequence of reductions:

1. **Transport cancellation.** The leading-order nonlinear contribution to the H^s energy evolution vanishes identically by incompressibility (equation (5.2)). This is an algebraic identity, not an estimate.

2. **Local stretching resolution.** The ACL bound, combined with Bernstein's inequality and the convergent series $\sum k_j^{3/2-s}$ for $s > 3$, absorbs the local stretching into the viscous dissipation (Proposition 5.3).

3. **KP-1 summability.** The KP-1 commutator contribution is controlled by Bernstein's inequality and the convergent series $\sum k_j^{5/2-s}$ for $s > 5/2$ (Lemma 5.5).

4. **KP-2 isolation.** After absorbing all other terms, the H^s energy evolution reduces to $dY/dt + \frac{3}{2}\nu Z \leq \sum |KP2_j|$ (Theorem 5.5). The regularity question reduces entirely to controlling the KP-2 sum.

5. **Trace-free geometry.** The sweeper strain tensor is symmetric and trace-free (by incompressibility). Contraction with an isotropic shell energy distribution gives zero. The KP-2 coupling is therefore modulated by the shell anisotropy factor $\mathcal{A}_j$.

6. **The Catch-22.** Case 1 ($\mathcal{A}_j \leq \theta_j$): the k_j^2 from the commutator cancels exactly against the k_j^{-2} from the anisotropy threshold. Summing over shells and applying the energy identity produces a linear bound λY.

   Case 2 ($\mathcal{A}_j > \theta_j$): the threshold condition forces high wavenumber. The definitional bound $E_j \leq Y k_j^{-2s}$ and the convergent series $\sum k_j^{2-2s}$ for $s > 3$ produce a subcritical bound $\mu Y^{2-s/2}$.

7. **Subcritical closure.** The combined energy inequality $dY/dt \leq \lambda Y + \mu Y^{2-s/2}$ has global solutions because the subcritical term grows strictly slower than the linear term for large Y. The comparison principle gives $Y(T) \leq C(s, \nu, Y_0, E_0, T) < \infty$ for every finite T.

The two-derivative gap is not bridged by gaining derivatives. It is circumvented by decomposing the dangerous term into two complementary regimes, each controlled by a different mechanism, neither of which requires closing the derivative count.

4 The Biot–Savart Kernel and Euclidean Geometry

4.1 What Was Required

The follow-up work to Tao's barrier established that even preserving the vorticity equation, circulation theorem, helicity conservation, and transport of vortex lines is insufficient to prevent blowup if the Biot–Savart law is modified. The irreducible structural feature is therefore the specific form of the Biot–Savart kernel, which encodes Euclidean geometry through the relationship $u = (-\Delta)^{-1}(\nabla \times \omega)$.

Any successful regularity mechanism must ultimately derive its power from this kernel.

4.2 What Was Addressed

The Leray projector $\mathcal{P}_k = I - \hat{k} \otimes \hat{k}$ is the Fourier-space manifestation of the Biot–Savart law. It appears in every stage of the Angular Cancellation programme:

1. **In the ACL.** Factor A of the geometric decomposition (Lemma 4.20) uses $\hat{u}(p) \perp p$ to rewrite the advective coupling $\hat{u}(p) \cdot q$ as $\hat{u}(p) \cdot k$. This identity is a direct consequence of the Leray projector's orthogonality constraint.

   Factor B uses the specific quadratic dependence of $\mathcal{P}_q$ on $\hat{q}$ to produce the first-harmonic azimuthal oscillation. The projection $\mathcal{P}_q \xi_k$ depends on $\cos\phi$ and $\sin\phi$ through the components of $\hat{q}(\phi)$, with the orthogonality constraint $\xi_q \cdot \hat{q} = 0$ reducing the trigonometric degree by one, eliminating the constant term.

2. **In the trace-free geometry.** The sweeper strain tensor $S_{ij}(p) = \frac{1}{2}(p_i \hat{u}_j(p) + p_j \hat{u}_i(p))$ is trace-free because $\mathrm{Tr}(S) = p \cdot \hat{u}(p) = 0$ by incompressibility. This trace-free property is equivalent to the statement that the Biot–Savart kernel produces a divergence-free velocity field: the strain of a divergence-free field is necessarily trace-free.

3. **In the collinear starvation.** The advective coupling $(\hat{u}(p) \cdot k)^2 \propto \sin^2 \alpha$ vanishes when $p \| k$ because $\hat{u}(p) \perp p$ and $p \| k$ forces $\hat{u}(p) \perp k$. This is the Biot–Savart constraint in its most direct form: the velocity generated by a vortex filament is perpendicular to the filament direction.

4. **In the $\sin(2\alpha)$ commutator vanishing.** The KP-2 commutator coupling vanishes at both geometric extremes ($\alpha = 0$ and $\alpha = \pi/2$) because the Biot–Savart kernel forces the coupling to factor through $\sin(2\alpha)$, which has zeros at both endpoints.

The Leray projector is not an auxiliary tool in this programme. It is the primary source of every geometric cancellation. The programme derives its power from the Biot–Savart kernel exactly as the barrier analysis requires.

5 Cascade Isotropisation and Turbulent Self-Attenuation

5.1 What Was Required

Computational evidence demonstrates that extreme vorticity events in Navier–Stokes turbulence exhibit self-attenuation: locally induced stretching becomes negative near the most intense vortex structures, preventing unbounded growth. The mechanism has been described as "twisting vortex lines regularise Navier–Stokes turbulence."

The research question: does this observed self-attenuation reflect a provable geometric mechanism, or does it merely delay blowup without preventing it?

5.2 What Was Addressed

The cascade isotropisation mechanism provides the mathematical engine behind the observed self-attenuation.

1. **The reduction factor hierarchy.** When the ACL-bounded cascade transfers energy between shells, the angular harmonic content of the energy distribution is filtered. The monopole ($\ell = 0$, total energy) passes unchanged: $R_0 = 1$. The quadrupole ($\ell = 2$, leading anisotropy) is reduced by $R_2 = -52/203$. The hexadecapole ($\ell = 4$) is reduced by $R_4 = 479/9744$. These are exact rational numbers computed from polynomial integrals.

2. **The spectral gap.** The ratio $|R_4|/|R_2| \approx 0.192$ establishes a spectral gap of factor ~ 5.2 between successive angular harmonics. Higher harmonics are damped more aggressively by the cascade. The load-bearing conditions $|R_2| < 1$ and $|R_4| < |R_2|$ are verified by exact integer arithmetic: $479 \times 203 < 52 \times 9744$.

3. **The Gaunt coupling structure.** The Navier–Stokes nonlinearity couples different angular sectors through Gaunt (Clebsch–Gordan) coefficients. The sweeper creates only $\ell = 2$ anisotropy (because the strain tensor is rank-2 symmetric traceless). Higher harmonics ($\ell = 4, 6, \ldots$) are created by self-interaction of lower harmonics and are subordinated to the quadrupole: $|a_{40}| \leq C|a_{20}|^2/E_j$.

4. **The effective quadrupolar ODE.** The anisotropy $a_{20}^{(j)}$ satisfies an effective damped ODE with cascade damping rate $\Gamma_2^{(j)} = 2\nu k_j^2 + (255/203)C_{\mathrm{ACL}} k_j^{7/2} E_j^{1/2}$, sweeper source $|S_j| \leq CG(t)E_j$, and cubic cross-sector correction from the Gaunt hierarchy.

5. **The round-trip contractivity.** The dominant leakage pathway $a_{20} \to a_{40} \to a_{20}$ through the $(2,2) \to 4$ and $(4,2) \to 2$ Gaunt channels has round-trip gain $\eta \approx 0.78 < 1$. The cross-sector coupling is contractive: it actively suppresses the cubic correction rather than amplifying it.

The cascade isotropisation is the mathematical proof that the computationally observed self-attenuation is a genuine geometric mechanism of the incompressible Navier–Stokes equations, not an artifact of finite resolution or statistical averaging.

6 Geometric Regularity Criteria

6.1 What Was Required

The Constantin–Fefferman criterion (1993) establishes that regularity follows if the direction of vorticity $\omega/|\omega|$ is Lipschitz in regions of high $|\omega|$. This is a conditional geometric criterion: it identifies a sufficient condition for regularity without establishing that the condition is satisfied.

The sparseness framework reduces the supercriticality of the Navier–Stokes equations by approximately 40% through geometric measure theory on super-level sets of vorticity combined with harmonic measure maximum principles. This is the largest scaling reduction achieved since the 1960s.

6.2 What Was Addressed

The Angular Cancellation programme differs from both approaches in kind, not merely degree.

1. **Unconditional vs. conditional.** The Constantin–Fefferman criterion is conditional: regularity follows *if* vorticity direction is coherent. The ACL and the Catch-22 are unconditional: they hold for every divergence-free velocity field on $\mathbb{T}^3$ with no assumption on phase coherence, amplitude concentration, or vorticity direction.

2. **Complete closure vs. partial reduction.** The sparseness framework reduces the scaling gap by approximately 40%. The Angular Cancellation programme reduces the shell-local scaling weight from $\sigma = +1/2$ (supercritical) to $\sigma = 0$ (critical) at the energy level, and closes the H^s energy inequality entirely through the subcritical Catch-22 mechanism.

3. **Shell-local Fourier geometry vs. physical-space vorticity geometry.** The programme operates in Fourier space on the integer lattice, exploiting shell-local triadic structure. The Constantin–Fefferman and sparseness approaches operate in physical space, exploiting pointwise vorticity structure. The two frameworks address different geometric aspects of the same nonlinearity.

7 Formal Verification

7.1 What Was Required

The research community has identified a vast infrastructure gap between existing proof assistant libraries and the mathematics required for Navier–Stokes analysis. Completely absent from all proof assistants are: Fourier analysis on $\mathbb{T}^3$, Sobolev spaces and their embeddings, distribution theory, Calderón–Zygmund theory, Littlewood–Paley decomposition, and any PDE existence or regularity theory.

No PDE regularity result of any kind has ever been formally verified in any proof assistant.

The permanent-axioms approach — declaring classical results as axioms and building the logical structure on top — has been identified as the most feasible path. Coq's `Print Assumptions` command makes the gap between formalised and assumed content fully transparent.

7.2 What Was Addressed

The formalisation consists of three independent Coq files totalling approximately 59 Qed theorems with zero Admitted statements, verified through the trusted kernel checker `coqchk`.

1. **All novel mathematical content is machine-verified.** Every new theorem in the programme — the ACL bound, the trace-free cancellation, the Catch-22 case split, the k_j^2 algebraic cancellation, the subcritical energy inequality assembly, and the main theorem — terminates in Qed with zero admitted statements.

2. **All axiomatised content is classical.** The permanent axioms encode textbook results: Sobolev embedding, Kato–Ponce commutator estimates, Calderón–Zygmund, energy identity, continuation criterion, parabolic regularity, and weak–strong uniqueness. These are axiomatised because MathComp lacks the infrastructure, not because they are uncertain.

3. **The trust boundary is documented.** The FIELDS-COQ Protocol (Version 5.0) provides six verification layers: structural anchor, anti-vacuity guard, atomic mapping appendix, residual audit, contradiction register, and hypothesis-to-axiom pipeline. Every hypothesis is classified as Proved, Dischargeable, or Permanent Axiom.

4. **The contradiction history is preserved.** The algebraically contradictory `catch_case2` hypothesis (28 February 2026) is permanently documented: which systems missed it, which caught it, and how it was replaced. The Contradiction Register (F-C 5.2) ensures that no error is erased from the record.

5. **The modular architecture is deliberate.** Interface hypotheses connect the three files without import dependencies. The architecture prevents file-level entanglement and allows each file to be compiled and verified independently.

This constitutes the first machine-verified theorem about the geometric structure of Navier–Stokes triadic interactions, and the first application of a structured governance protocol to AI-assisted formal mathematics.

8 Energy-Level and H^s-Level Coverage

8.1 What Was Required

The energy identity provides global control at the L^2 level. The regularity problem requires control at H^s for s sufficiently large. The standard continuation criterion requires $\sup_{[0,T^*)} \|u\|_{H^s} < \infty$ for some $s > 5/2$.

8.2 What Was Addressed

The programme provides both energy-level and H^s-level results:

Component	Level	Status
ACL bound	Energy (1 derivative)	Machine-verified
Cascade locality	Energy	Proved
Transport cancellation	H^s (algebraic identity)	Proved
Local stretching resolution	H^s for $s > 3$	Proved
KP-1 summability	H^s for $s > 5/2$	Proved
KP-2 isolation	H^s	Machine-verified
Trace-free cancellation	H^s	Machine-verified
Catch-22 Case 1	H^s	Machine-verified
Catch-22 Case 2	H^s for $s > 3$	Axiomatised
Subcritical closure	H^s for $s > 3$	Machine-verified
Continuation	H^s for $s > 5/2$	Axiomatised
Pressure control	H^{s-1}	Machine-verified
Uniqueness	L^2	Axiomatised
Continuous dependence	L^2 and $H^{s'}$, $s' < s$	Axiomatised
Parabolic smoothing	C^∞ for $t > 0$	Axiomatised

9 The Vorticity Equation and Helicity Structure

9.1 What Was Required

The vorticity formulation $\partial_t \omega + (u \cdot \nabla)\omega = (\omega \cdot \nabla)u + \nu \Delta \omega$ is identified as the most promising structural feature absent from Tao's averaged system. The vortex stretching term $(\omega \cdot \nabla)u$ is the mechanism by which enstrophy can grow, and controlling it is equivalent to the regularity problem.

The helical decomposition reveals that interactions between same-sign helicity modes vanish in certain configurations, reducing the effective number of active triads.

9.2 What Was Addressed

The programme does not work directly in the vorticity formulation. Instead, it controls vortex stretching indirectly through the Fourier-space triadic decomposition:

1. **Shell-local vortex stretching.** The quantity $VS_j = \int_{\mathbb{T}^3} \omega_j \cdot S[u_j]\omega_j \, dx$ is the restriction of vortex stretching to shell S_j. The ACL bounds this quantity directly.

2. **The Bernstein connection.** On a unit-width shell, $|\hat{\omega}(k)| \leq |k||\hat{u}(k)|$ (pointwise inequality). This connects the Fourier-space coupling bounds to the physical-space vortex stretching integral.

3. **Anisotropy as a proxy for vorticity coherence.** The quadrupolar anisotropy $\mathcal{A}_j$ measures the angular concentration of Fourier energy on shell S_j. High anisotropy corresponds to directionally coherent vorticity (the physical-space condition that Constantin–Fefferman identified as dangerous). The Catch-22 controls this quantity: either $\mathcal{A}_j$ is

small (cascade-dominated, k_j^2 cancellation applies) or $\mathcal{A}_j$ is large but the shell has high wavenumber (sweeper-dominated, subcritical summability applies).

The helicity structure is not explicitly exploited. Remark 4.31 in the manuscript notes that helicity decomposition provides constant-factor suppression (same-helicity cancellation) but no exponent improvement. The programme achieves its exponent improvement through the transverse-collinear decomposition instead.

10 Domain and Boundary Considerations

10.1 What Was Required

The Clay Millennium Problem accepts solutions on either $\mathbb{R}^3$ or $\mathbb{T}^3$. The Chen–Hou Euler blowup result relies on a cylindrical boundary to stabilise the singularity. Extension to the whole-space or periodic setting remains open.

10.2 What Was Addressed

The programme works exclusively on the periodic torus $\mathbb{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3$. This choice is not a limitation but a structural requirement:

1. **Discrete spectrum.** The Fourier spectrum on $\mathbb{T}^3$ is the integer lattice $\mathbb{Z}^3$, which enables the lattice counting arguments (shell cardinality, resonant set geometry, collinear tube bounds) that drive the ACL.
2. **No boundary terms.** Periodicity eliminates all boundary contributions in integration by parts. The energy identity, the transport cancellation, and the Kato–Ponce commutator structure all rely on this.
3. **Poincaré inequality.** The Poincaré inequality $\|f\|_{L^2} \leq k_0^{-1}\|\nabla f\|_{L^2}$ is available on $\mathbb{T}^3$ but not on $\mathbb{R}^3$. It provides the lower bound $P \geq k_0^2\Omega$ that connects palinstrophy to enstrophy.
4. **Compact domain.** No spatial decay hypotheses are needed. The function spaces $H^s(\mathbb{T}^3)$ are compactly embedded for $s' < s$, simplifying the functional framework.

Extension to $\mathbb{R}^3$ would require replacing the lattice counting arguments with continuum analogues. This is identified as future work.

11 Non-Uniqueness and the Solution Landscape

11.1 What Was Required

Recent results have expanded the non-uniqueness programme: non-uniqueness of Leray solutions for forced Navier–Stokes, and non-uniqueness of smooth solutions from critical data. These results reveal that the solution landscape is more complex than previously understood.

11.2 What Was Addressed

The programme establishes weak–strong uniqueness (Theorem 8.1): the global strong solution constructed in Theorem 6.23 is the unique Leray–Hopf weak solution with the same initial data. This is the standard uniqueness result for strong solutions and is unaffected by the non-uniqueness phenomena, which concern either forced equations or critical (not subcritical) data.

The initial data hypothesis $u_0 \in H^s$ with $s > 3$ places the data well above the critical regularity $H^{1/2}$. No smallness condition is imposed.

12 Claimed Proofs and Community Scrutiny

12.1 What Was Required

The research community maintains a high threshold for claimed proofs of Navier–Stokes regularity. No claimed proof has survived expert scrutiny. The standard expectation is that any valid proof would require either discovering new geometric structure or constructing an explicit blowup solution.

12.2 What Was Addressed

The manuscript's approach to community scrutiny includes:

1. **Machine verification.** The core argument is formalised in Coq and verified through the trusted kernel checker. This provides a verification standard independent of human referee judgment.

2. **Explicit scope limitation.** The original ACL paper (Sections 1–4) explicitly states that it does not claim global regularity. The two-derivative enstrophy gap is identified as the remaining barrier. The extension (Sections 5–9) addresses this gap through the cascade isotropisation and subcritical closure.

3. **Contradiction transparency.** The `catch_case2` contradiction is permanently documented. The discovery, the failure of multiple systems to detect it, and the replacement mechanism are all recorded in the Contradiction Register.

4. **Audit infrastructure.** The FIELDS Protocol, the FIELDS-COQ Protocol, and the Common Failure Modes documentation provide a reproducible framework for independent evaluation.

5. **Explicit constant ledger.** Every constant in the proof is recorded with its definition, first use, dependence chain, and numerical value (Appendix A). No constant is chosen adaptively.

13 Summary of Coverage

Requirement	Status	Reference
Exploit structure absent from Tao's averaged system	Addressed	ACL, §4.5
Use Biot–Savart / Leray projector geometry	Addressed	§4.5, §5.5
Lattice geometry of $\mathbb{Z}^3$	Addressed	§4.5, App. G.7
Close the two-derivative gap	Circumvented via structural decomposition	§5–6
Cascade isotropisation	Addressed	§6.2–6.5
Gaunt coupling control	Addressed	§6.3, App. G
Spectral gap in reduction factors	Verified by exact arithmetic	Thm. 6.2, App. D.4.7
Subcritical energy closure	Addressed	Prop. 6.21
Continuation criterion	Applied (axiomatised)	Thm. 9.1
Pressure reconstruction	Addressed	§7
Weak–strong uniqueness	Applied (axiomatised)	Thm. 8.1
Continuous dependence	Addressed	Thm. 8.2
Parabolic smoothing	Applied (axiomatised)	Cor. 9.5
Machine verification	59 Qed, 0 Admitted, 3 files, `coqchk`	App. H
Contradiction transparency	Documented	App. H.8
Trust boundary documentation	FIELDS-COQ Protocol	App. H
Audit methodology	FIELDS Protocol + Common Failure Modes	Separate documents

14 Open Items

The following items are explicitly identified as outside the scope of the present programme:

1. **Extension to $\mathbb{R}^3$.** The lattice counting arguments are specific to $\mathbb{T}^3$. Continuum analogues would require different geometric tools.

2. **Full Coq formalisation of the Case 2 subcritical bound.** This requires fractional exponents and series convergence infrastructure not available in MathComp. Axiomatised as a permanent axiom.

3. **Full Coq formalisation of the Duke–Linnik equidistribution theorem.** Identified as the most technically demanding future formalisation target (Remark G.28).

4. **Full Coq formalisation of all PDE infrastructure.** Sobolev embedding, Kato–Ponce, Calderón–Zygmund, continuation criterion, and parabolic regularity are axiomatised. Decades of library development would be required to formalise these.

5. **Helicity decomposition.** Provides constant-factor suppression but no exponent improvement. Not exploited in the current programme.

6. **Optimal constants.** The ACL constant $C_{\mathrm{ACL}} \leq 8\pi^3 c_0^{-1/2}$ and the Gronwall coefficients λ, μ are explicit but not optimised. Sharper constants would improve the quantitative bound but do not affect the qualitative result.

15 Closing Note

This pamphlet does not claim that the Angular Cancellation programme resolves the Navier–Stokes Millennium Problem. It documents, in systematic detail, the technical requirements identified by the research community, and traces how each requirement was addressed or scoped within the programme.

The mathematics speaks for itself. The reader is invited to evaluate the coverage, trace the logical chain, compile the Coq files, and form an independent judgment.

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